

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Student Assessment Document

FINAL EXPLANATION: GENERAL SCIENCE GRADE 10 – SUB-STRAND 2.4 ACIDS, BASES AND SALTS: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Look again at the two images from Lesson 1: (1) Lake Magadi with its white trona crusts and pink-red alkaline water, and (2) the aluminium sufuria with white scale inside and pitting outside where lemon-water sat overnight. Use everything you learned in Lessons 1 to 8 to answer each question in the space under it. Write in full sentences. Use the Claim–Evidence–Reasoning (CER) pattern where the question asks for it: state your claim, give evidence from our investigations, and explain your reasoning using ions, particles or reactions. Include equations or sketches where they help your answer. Use correct chemical vocabulary and Kenyan examples from our lessons.

Part 1: Why is Lake Magadi water caustic, and why does it form white crusts?

Lake Magadi water looks dangerously caustic, yet the Maasai have harvested its salts for centuries and the Magadi Soda Company makes soda ash from it. Using the Claim–Evidence–Reasoning pattern, explain (a) why the water is caustic, in terms of pH and ions, and (b) how the white trona crusts form on the shoreline. Name the salt in the crusts and the product made from it.

Write your answer here:

Part 2: Why did lemon juice pit the sufuria?

Lemon juice is safe to drink, yet lemon-water left overnight pitted the outside of the aluminium sufuria. Explain this using the pH of lemon juice, the ions involved, and what happened when we reacted an acid with a metal in Lesson 4. Write a word equation for the type of reaction. Then explain why the same lemon juice does not damage your body when you drink it.

Write your answer here:

Part 3: Why does salt clump, and where did the white scale come from?

(a) A school cook in Mombasa finds her bulk salt clumps into a block in the rainy season. Using our petri dish results for NaCl , CaCl_2 , $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ and $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, explain why this happens. Use the terms hygroscopic, deliquescent and efflorescent correctly. (b) The white scale inside the sufuria is also a salt. Explain how it formed, and say how it differs from the trona at Lake Magadi.

Write your answer here:

Part 4: Salts in our lives and the environment

Answer both parts. (a) Choose TWO different sectors from Lesson 6 (agriculture, food, medicine, laundry, road use). For each, name a specific salt and explain how its chemistry makes it useful. (b) Using the Lake Victoria case study from Lesson 8, explain how using too much fertiliser salt can harm the lake and its fish. Then answer: how can understanding salt chemistry help us use salts safely and sustainably?

Write your answer here:

Part 5: Your model has changed

Think back to your Lesson 1 two-panel sketch and predictions. (a) Describe one idea you held in Lesson 1 that you have now changed or improved, and say which evidence caused the change. (b) Draw and label a particle-level sketch comparing the ions in lemon juice with the ions in Lake Magadi water. (c) Answer the driving question in two to three sentences: why do these substances react so differently with metals, food and living things?

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of evidence from the investigations	Cites specific results from the lessons (pH ranges, Mg + HCl observations, petri dish results, nitrate case study) for every claim.	Cites some specific evidence but some claims rely on general statements.	Gives little or no evidence from the investigations, or evidence is inaccurate.
Particle-level reasoning (ions)	Consistently explains phenomena using H^+ , OH^- , Na^+ , Cl^- and ion–water attraction, accurately linked to pH, reactions and salt behaviour.	Uses ions in most explanations, with minor gaps or errors.	Describes only what is seen, with little or incorrect use of ions.
Chemical vocabulary and equations	Uses terms such as hygroscopic, deliquescent, efflorescent, neutralisation and water of crystallisation correctly, and writes accurate word equations.	Uses most terms correctly; word equations are mostly correct.	Terms are missing or misused; equations are missing or incorrect.
Connection to Kenyan contexts and the anchoring phenomenon	Fully explains both Lake Magadi and the sufuria, and links salt applications and Lake Victoria impacts to the driving question.	Explains both images, with limited links to applications or the environment.	Explains only one image or gives disconnected examples.
Reflection and model revision	Clearly identifies a changed idea, the evidence that changed it, and gives an accurate labelled particle sketch and a coherent answer to the driving question.	Identifies a changed idea with some evidence; sketch and final answer are mostly accurate.	Reflection is vague; sketch or final answer is missing or incorrect.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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